

CompTIA A+ Core 1 Playbook

IT Labs

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7.3.3 Fix a Network Connection

You are the IT administrator for a small corporate network. The person using the Office1 computer has called you and said that they can no longer access the network or the internet. You suspect that the user has been adjusting their computer on their own.

In this lab, your task is to diagnose and fix the problem.

You will know you have fixed all the problems when the Office1 computer can ping the Office2 computer and ping the DNS server.

While troubleshooting this issue, consider the following:

- The network does not have a DHCP server.
- The default subnet mask should be used for all computers.
- The connection problem might be in the operating system configuration, the hardware, or both.
- There might be more than one problem.
- During the lab, the Diagnose and repair option is not available. Instead, review the configuration of the system to identify the problem. You can view the connection properties or use the **ipconfig** command.
- Use the information in the Exhibit to verify the required network connection settings.
- Use the Settings app's Network & internet settings to help you verify whether you have fixed the problem(s).

IP addresses below .15 are being used by the servers. IP addresses between .30 and .34 are being used by other workstations.

Explanation

Complete this lab as follows:

1. Verify the problem.
 1. From the notification area, right-click the **network icon** and select **Network & Internet settings**.

Notice that you are shown that the computer has no connection to the internet.
 2. Select **Advanced network settings**.
 3. Under *Related Settings*, select **More network adapter options** to help determine why.

Notice the status for the network connection is *Network cable unplugged*.
 4. Close the Network Connections window.
 5. Close the Settings app.
2. Connect the cable.
 1. From the top left, select **Office 1** to work with the computer hardware.
 2. Switch to the **Back** view of the computer.

The computer has a network cable, but the link light is not on.
 3. Select the **network cable** connected to the computer to see what is connected to the other end.

Notice that in the Selected Item pane, one end is unconnected.
 4. Select the **unconnected end** and drag it to the network connection on the wall plate.
3. Configure the IP addresses.
 1. Select the monitor to switch to the operating system view.

The notification area shows a network connection.
 2. Right-click **Start** and select **Settings**.
 3. Select **Network & internet**.
 4. Select **Ethernet**.
 5. For *IP assignment*, select **Edit**.
 6. If necessary, change the Edit IP settings dropdown to **Manual**.
 7. For IPv4, make sure the button is slid to the **ON** position.
 8. Configure the fields as follows:
 - IP address: **Valid IP address** (such as 192.168.0.27)
 - Subnet mask: **255.255.255.0**
 - Gateway: **192.168.0.5**
 - Preferred DNS: **163.128.78.93** or **163.128.80.93**

9. Select **Save**.
 10. Close the Settings app.
4. Verify a connection to the internet.
 1. Right-click **Start** and select **Terminal**.
 2. Type **ping 163.128.78.93** from the PowerShell prompt.
 3. Press **Enter**.